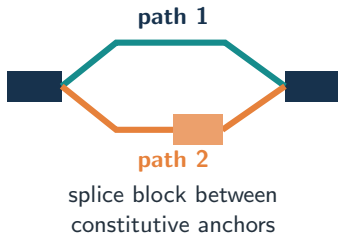


Differential splicing from equivalence-class counts

Local splice-path effects propagated through isoforms and
primer-aware observations

The target is a local splice choice with biological replication



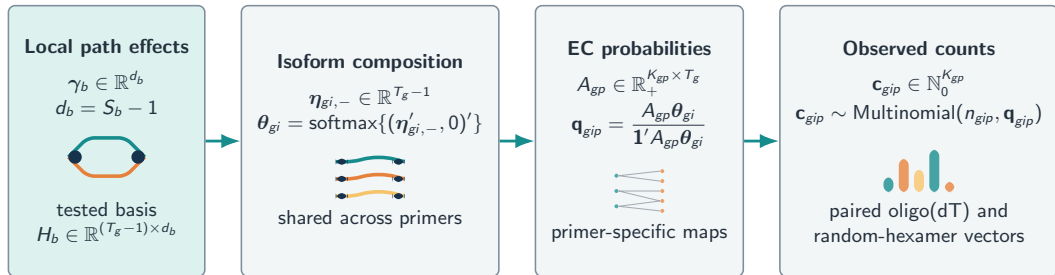
Two questions

Cell type, does path usage differ across cell types while condition is unrestricted?

Condition, does path usage differ across conditions within one cell type?

- ▶ One hypothesis per nonredundant local path partition
- ▶ Mouse random effects represent biological replication
- ▶ Cells contribute to mouse $\times$ condition $\times$ cell-type pseudobulks, never independent replicates

Forward model, splice-path effects $\rightarrow$ isoforms $\rightarrow$ ECs $\rightarrow$ counts



Ambiguous ECs remain ambiguous, primer design enters through A_{gp} , and a pooled-baseline Jacobian defines the local path effect without discarding global isoform nuisance variation.

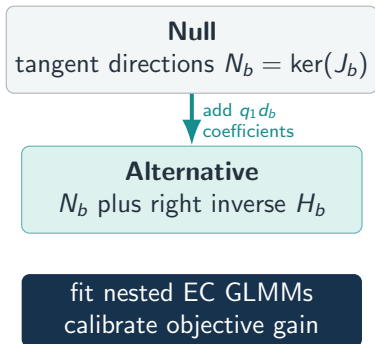
The corrected null follows constant local path usage

Isoform-logit predictor

A label-independent pooled fit defines the path-ILR Jacobian $J_b \in \mathbb{R}^{d_b \times (T_g - 1)}$. The tested basis is a right inverse, $J_b H_b = I_{d_b}$, and the nuisance basis spans its kernel, $J_b N_b = 0$.

The null predictor is $\mathcal{F}_{bi}^{(0)} = B_0' Z_{bi}' + \sum_j W_{bij} N_b \alpha_{bj}$, while the alternative adds $\sum_j W_{bij} H_b \gamma_{bj}$.

- ▶ $Z_b \in \mathbb{R}^{l_b \times q_0}$ carries nuisance covariates
- ▶ $W_b \in \mathbb{R}^{l_b \times q_1}$ carries the tested cell-type or condition design
- ▶ The alternative adds $q_1 d_b$ local path coefficients and remains unrestricted; for a pairwise cassette test, $d_b = 1$



Calibration matters

The ordinary multilevel chi-square LRT is anti-conservative under structured nulls. The primary mouse result uses simulated nulls, cross-fitted tail modeling, leave-block-out calibration, and BH.

Two brain datasets, one subject-level analysis design

Mouse

GSE189033

Biology, 5xFAD amyloidosis crossed with Csf1r Δ FIRE microglial deficiency, including PBS and microglia-transplant arms

Samples, 48 mice, six groups, balanced 4 female and 4 male per group, age 5–6 months

Assay, SPLiT-seq single-nucleus RNA-seq, paired oligo(dT) and random-hexamer observations

Analysis strata, 936 mouse $\times$ condition $\times$ cell-type pseudobulks before feature-level support filtering

Human

GSE233208

Biology, cortical postmortem tissue spanning control, sporadic Alzheimer disease, Down syndrome, and Down syndrome with AD

Scale, 396,277 annotated nuclei represented by 40 paired sequencing runs in five batches

Assay, Parse Biosciences Evercode WT v1 single-nucleus RNA-seq

Role, independent cross-species sensitivity analysis with the same EC GLMM fitting procedure

Sources, Spangenberg et al., Cell Reports 2022, GEO GSE189033; Miyoshi et al., Nature Genetics 2024, GEO GSE233208. Counts shown are the analysis metadata summarized in this repository.

Corrected full-data calibration yields no BH discoveries

522

nominal cell-type tests

0

cell-type BH calls

922

nominal condition tests

0

condition BH calls

- ▶ 4,825 cell-type path partitions and 14,833 condition-within-cell-type tests
- ▶ Aggregate null rejection matches 0.05, 0.01, and 0.001 after cross-fitted calibration
- ▶ Minimum calibrated p-values are 0.00216 for cell type and 0.000148 for condition, above their first BH thresholds
- ▶ The primary logistic-normal likelihood allows pseudobulk extra-multinomial variation shared across the two primer halves

Interpretation

The corrected tangent null removes the former-basis 238-call set. The earlier structural and literature catalog is hypothesis-generating rather than a validated discovery list.

Direct local-path Tealeaf exceeds junction comparators

Comparator	Tealeaf			Comparator		
	Rep.	held	ρ	Rep.	held	ρ
LeafCutter	50	0.799	0.711	26	0.750	0.542
MAJIQ Heterogen	103	0.810	0.622	38	0.698	0.437
scQuint	101	0.819	0.656	56	0.775	0.458

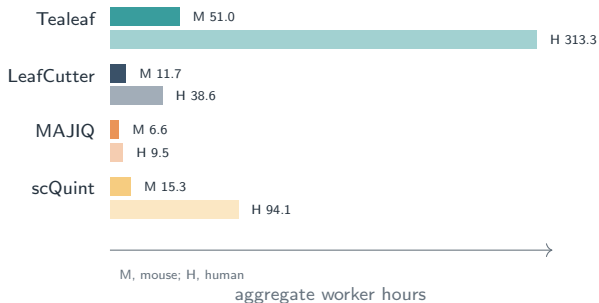
What the paired local-path test changes

Subject-level path ILRs are estimated directly from both primer-specific EC vectors, stabilized by add-two smoothing, variance-moderated across scalar tests, and empirically calibrated with 32 sign-flip families. “Rep.” applies BH to the conjunction p-value on each pairwise shared gene universe; ρ is fold correlation of gene-pair $-\log_{10} p$.

Interpretation

Every aggregated replicated-gene null family has zero BH calls. Add-two smoothing was selected on these splits, so independent-data confirmation remains necessary.

EC ambiguity and latent-Gaussian fitting carry a compute cost

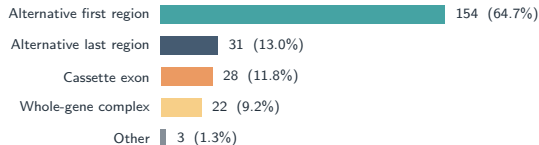


3.3×
slower than scQuint
broad worker time

- ▶ Same scalar Laplace LRT and analytic EC derivatives in both species
- ▶ Human has 4.3× more tests
- ▶ Median human test has 5 isoforms, 26 ECs, and 5 paths; mouse has 3, 9, and 2
- ▶ Peak memory, 4.81 GiB mouse and 8.25 GiB human

Worker time starts from prepared inputs and excludes shared read processing.

Robust mouse calls are enriched for transcript-end differences



Functional precedents among robust calls

- ▶ *Ntrk2*, alternative last region separating short kinase-deficient and long TrkB paths; TrkB.T1 has a characterized astrocyte role
- ▶ *Grin1*, C1 cassette exon linked to receptor surface trafficking
- ▶ *App*, exon 15 cassette linked to polarized secretion and amyloid processing
- ▶ *Enah*, exon 11a cassette linked to actin dynamics outside brain

Boundary of the annotation

A terminal-region label means supported paths differ between a constitutive anchor and a transcript boundary. It does not by itself distinguish promoter choice, terminal exon choice, or polyadenylation.

Takeaways

Model the observations that were measured. Local path effects act through full isoforms, primer-specific EC maps, and observed pseudobulk counts without hard EC allocation.

Separate biology from calibration. The corrected full-data EC GLMM has no BH discoveries; the former-basis functional catalog is exploratory.

Local inference realizes the power advantage. The calibrated paired local-path test has more replicated genes and higher held-half replication than every comparator.

The remaining tradeoff is compute. Retaining EC ambiguity and repeatedly solving nested latent-Gaussian models costs about $3.3\times$ scQuint's broad worker time after analytic derivative speedups.